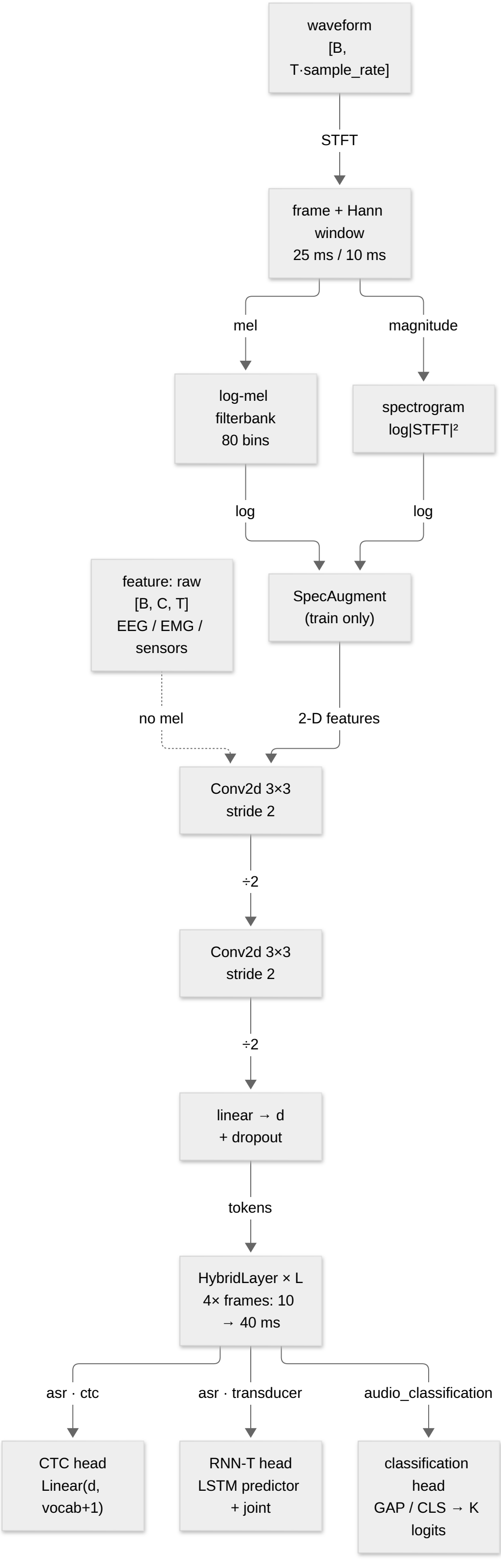

waveform
[B,
T·sample_rate]
STFT
frame + Hann
window
25 ms / 10 ms
mel
magnitude
log-mel
filterbank
80 bins
spectrogram
log|STFT|²
log
log
feature: raw
[B, C, T]
EEG / EMG /
sensors
SpecAugment
(train only)
no mel
2-D features
Conv2d 3×3
stride 2
÷2
Conv2d 3×3
stride 2
÷2
linear → d
+ dropout
tokens
HybridLayer × L
4× frames: 10
→ 40 ms
asr · ctc
asr · transducer
audio_classification
CTC head
Linear(d,
vocab+1)
RNN-T head
LSTM predictor
+ joint
classification
head
GAP / CLS → K
logits